COMP 5350/6350 Hands-On #3 Report

Student Name: Eristen Forsythe

1. Executive Summary

This investigation examined the network packet capture file FTPPackets.pcap to identify login activity and file transfers on the FTP server. The analysis showed that the authorized user **ftpuser** logged in and downloaded the file **employee_details.txt**, which contained employee usernames and passwords.

After this file was downloaded, additional logins were observed from the accounts **johnsmith** and **janedoe**, indicating that the credentials from the file were used to access the server. During this several files were uploaded and downloaded, including images, PDFs, and a ZIP archive. All transferred files were successfully reconstructed from the packet capture and verified using SHA-256 and MD5 hashes.

The recovered files included financial documents, check images, a mortgage template, and an archive containing identity-related images such as a social security card and birth certificate. These materials suggest that the unauthorized users were collecting or exchanging documents that could be used for identity theft or financial fraud.

Overall, the investigation shows that exposure of plaintext credentials in employee_details.txt enabled unauthorized access to the FTP server. The compromised accounts were then used to transfer several files associated with personal identity and financial documentation..

The timeline illustrates how the exposure of plaintext credentials enabled unauthorized access to the FTP server, which was then used to exchange financial and identity-related documents that could potentially be used for identity theft or financial fraud.

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Forensic Attack Timeline

Stage	Timestamp	User	Source IP	Activity Description	Evidence
1	15-11-2024 13:48:51	ftpuser	172.19.98.16	Authorized employee logs into the FTP server using valid credentials. This appears to be the legitimate initial access to the system.	FTP USER ftpuser and PASS commands followed by server response **230 Login Successful**
2	Shortly after login	ftpuser	172.19.98.16	The user downloads **employee_details.txt**, which contains employee usernames, passwords, and personal information. This file exposes valid FTP credentials for other accounts.	FTP command RETR employee_details.txt with responses **150 Opening data connection** and **226 Transfer complete**
3	15-11-2024 13:52:36	johnsmith	172.19.201.228	A new login occurs using the account **johnsmith** shortly after the credential file was downloaded, indicating that the exposed credentials were used to gain unauthorized access.	FTP USER johnsmith followed by successful authentication response **230 Login Successful**
4	After login	johnsmith	172.19.201.228	The unauthorized user uploads a completed check image and downloads a bank statement template, suggesting preparation or modification of financial documents.	FTP STOR Check1.jpg and RETR Statement2.pdf commands
5	15-11-2024 13:54:16	janedoe	131.204.27.177	Another account contained in the credential file (**janedoe**) logs in from a different IP address, indicating additional unauthorized access using exposed credentials.	FTP USER janedoe followed by server response **230 Login Successful**
6	After login	janedoe	131.204.27.177	The user downloads multiple files including a check template, bank statement, and a ZIP archive containing identity documents such as an SSN card and birth certificate.	FTP RETR check1.jpg, RETR Statement1.pdf, and RETR Documents.zip
7	Same session	janedoe	131.204.27.177	The attacker uploads a **Mortgage.pdf** template, which could be used alongside the collected financial and identity documents for fraudulent financial activity.	FTP STOR Mortgage.pdf observed in the packet capture
8	Final stage	Multiple	Multiple	The activity results in the collection and exchange of financial documents and identity records, indicating potential preparation for identity theft or financial fraud.	Recovered files include checks, bank statements, mortgage forms, and identity documents

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2. Investigation Methodology (General Overview)

The investigation was performed by analyzing a network packet capture file (**FTPPackets.pcap)** that contained recorded FTP traffic between several client machines and an FTP server. Before beginning the analysis, **hash values** were generated for the **PCAP** file to verify the integrity of the evidence.

The packet capture was examined using **Wireshark** on a **Kali Linux system**. Wireshark was used because it allows inspection of network packets, filtering by protocol, reconstruction of TCP streams, and exporting transferred files.

Manual analysis was performed by applying filters to isolate FTP traffic and identify authentication events and file transfers. Login attempts were identified by examining the FTP USER and PASS commands, and successful logins were confirmed using the FTP response code 230 (Login successful).

File transfers were identified by analyzing FTP commands such as RETR (download) and STOR (upload). These commands indicate when files are retrieved from or stored on the FTP server. The associated TCP streams were followed in Wireshark to reconstruct and export the transferred files.

Recovered files were saved locally and verified using hashing tools available in Kali Linux. The commands **sha256sum** and **md5sum** were used to generate hash values for each recovered file to verify file integrity.

Throughout the investigation, screenshots were captured to **document** key steps of the analysis process, including protocol statistics, login activity, FTP commands, and file recovery. These screenshots were annotated and included in the report to support the findings and ensure that the analysis process is transparent and repeatable.

Evidence file hashes were calculated prior to analysis:

MD5: 888af7dede8e3711d12eaafaf3881562

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SHA-1: 96bb76d0151c0c2b0079b1f2b77e1ca31ab8c757
SHA-256: d2692f0fa84546a48c5b26486f3ff0718b6bf7f9f57b0776b62116cac972d1cf

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3. Evidence Overview

3.1 PCAP File Properties

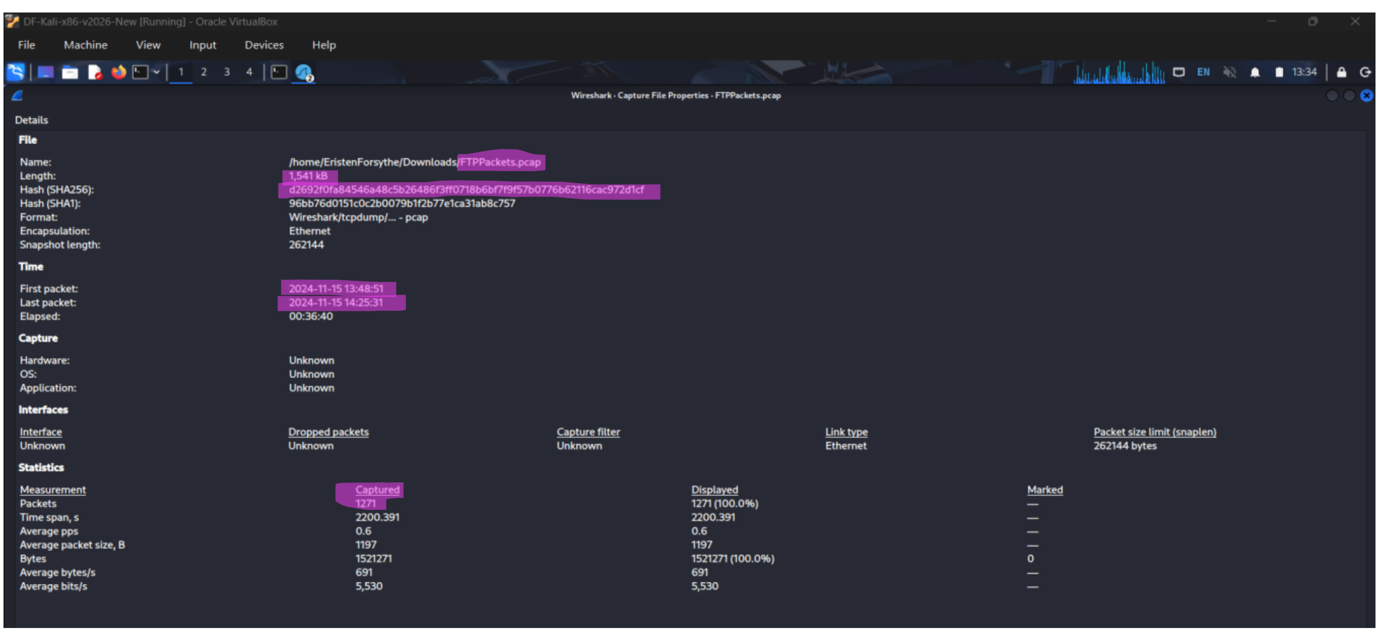
- File name: FTPPackets.pcap
- File size: 1,541 KB (1,541,631 bytes)
- Hash values (MD5/SHA-1/SHA-256)

MD5: 888af7dede8e3711d12eaafaf3881562

SHA-1: 96bb76d0151c0c2b0079b1f2b77e1ca31ab8c757

SHA-256: d2692f0fa84546a48c5b26486f3ff0718b6bf7f9f57b0776b62116cac972d1cf

- Total packets : 1271
- Protocol breakdown: FTP control and FTP data traffic
- Capture duration: 00:36:40



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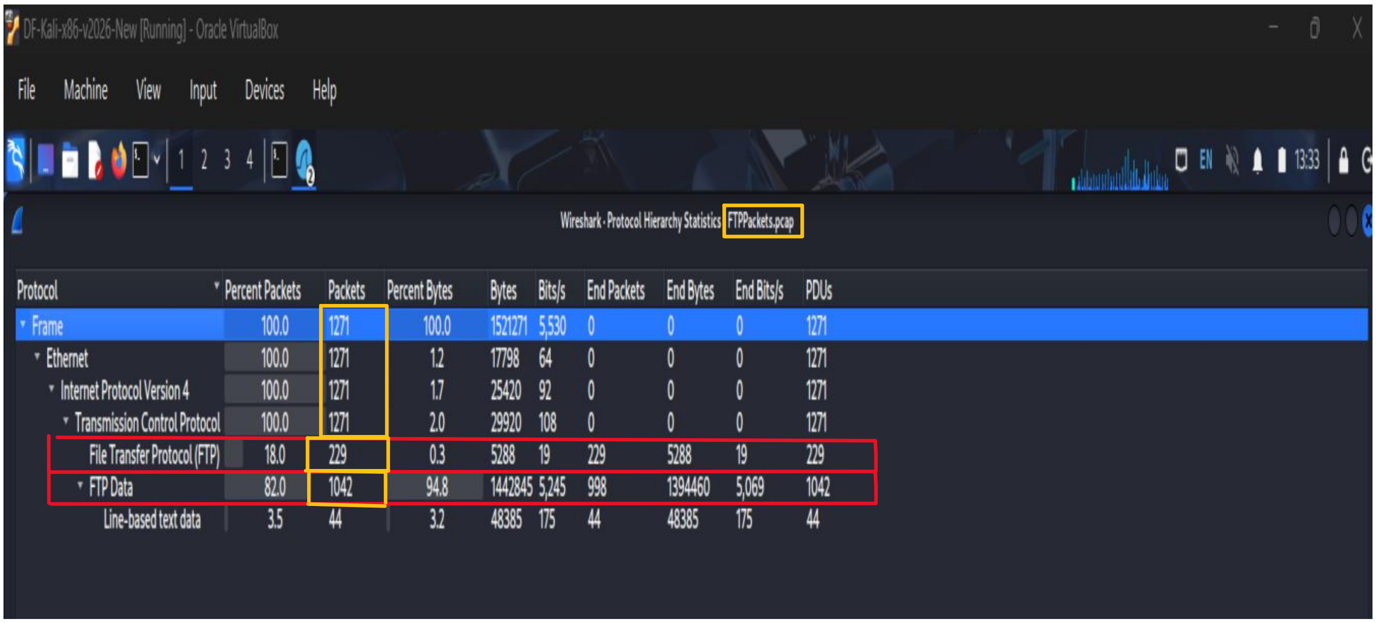
Wireshark capture file properties showing the evidence file name, capture timestamps, packet count, and SHA-256 hash used to check integrity.



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	1271	100.0	1521271	5,530	0	0	0	1271
Ethernet	100.0	1271	1.2	17798	64	0	0	0	1271
Internet Protocol Version 4	100.0	1271	1.7	25420	92	0	0	0	1271
Transmission Control Protocol	100.0	1271	2.0	29920	108	0	0	0	1271
File Transfer Protocol (FTP)	18.0	229	0.3	5288	19	229	5288	19	229
FTP Data	82.0	1042	94.8	1442845	5,245	998	1394460	5,069	1042
Line-based text data	3.5	44	3.2	48385	175	44	48385	175	44

Protocol hierarchy statistics showing FTP control traffic and FTP data transfers present in the packet capture

3.2 Network Environment Summary

- FTP Server IP address: 131.204.27.176
- Server MAC address: 1c:69:7a:9c:f8:6c
- Subnet involved: 172.19.0.0/16 and 131.204.27.0/24
- Observed client IP addresses: 172.19.98.16, 172.19.201.228, and 131.204.27.177
- Number of unique connections: 3

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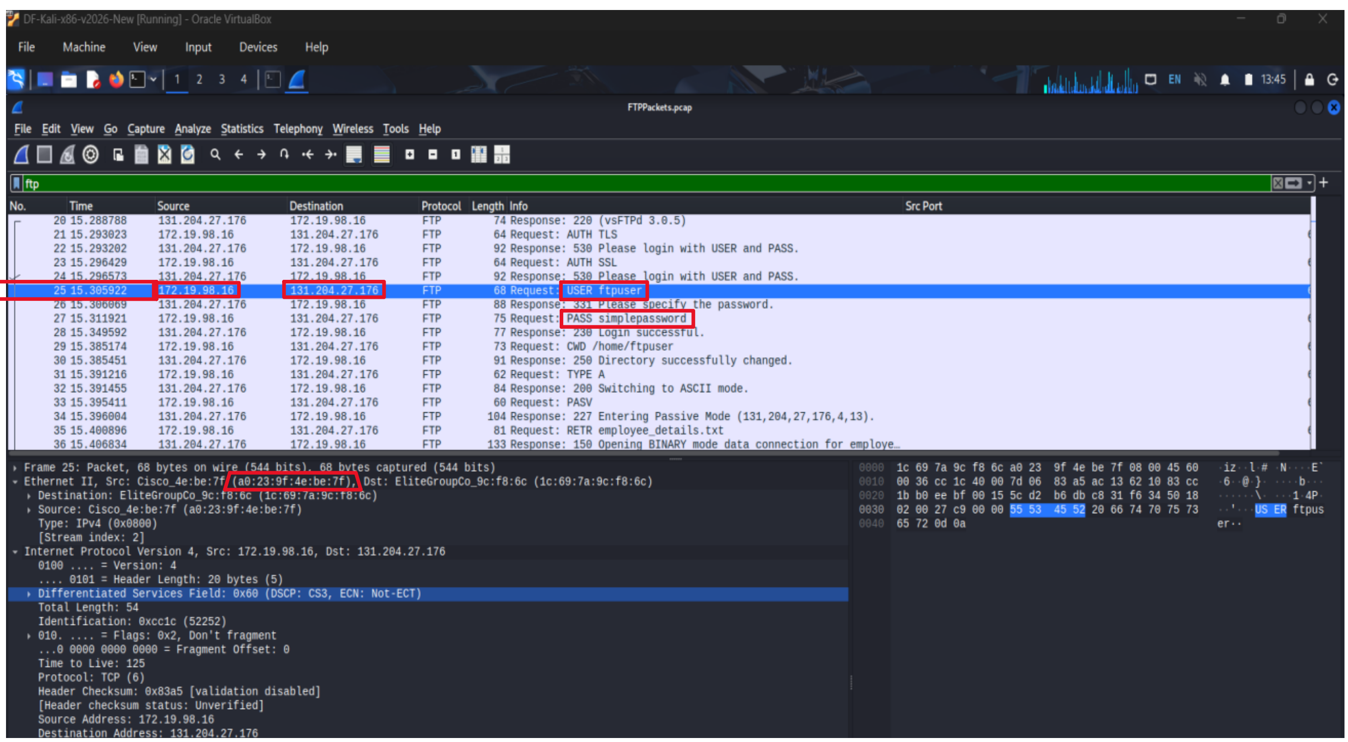
4. Authorized Login & Initial File Download

4.1 Authorized User Identification

- ftp_username: ftpuser
- Login Timestamp (DD-MM-YYYY HH:MM:SS): 15-11-2024 13:48:51
- Source IP address: 172.19.98.16
- MAC address: a0:23:9f:4e:be:7f
- FTP response codes: 331 (user name okay, but needs password) and 230 (login successful)

Include annotated screenshots of:

- USER/PASS commands
- Successful login confirmation



FTP authentication traffic showing the USER and PASS commands for the authorized account ftpuser, coming from IP address 172.19.98.16

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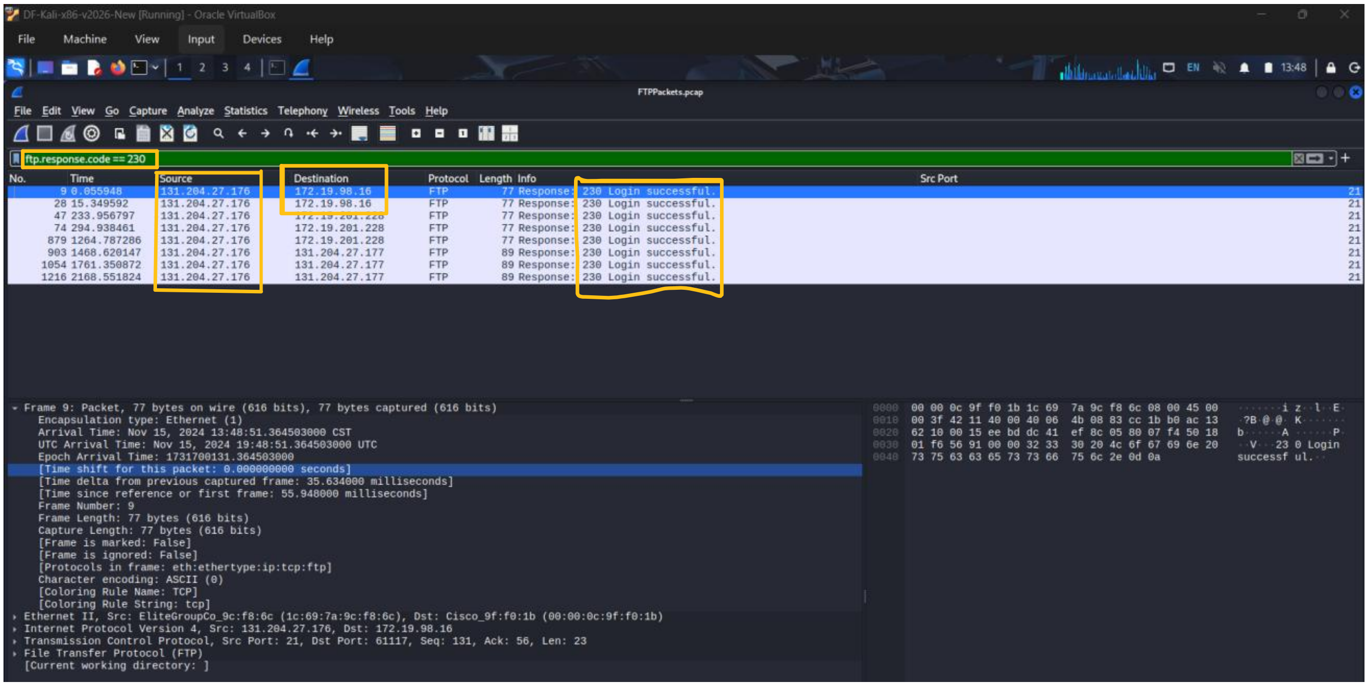

FTP server response 230 Login successful confirming authentication of the user ftpuser

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4.2 Initial File Download

- File name: employee_details.txt
- File size (bytes): 485 bytes
- Transfer confirmation (FTP 150/226 responses): 150 (opening data connection) and 226 (transfer complete)
- File reconstruction steps:
    - Located the FTP command RETR employee_details.txt in wireshark.
    - Right-clicked the packet and selected Follow -> TCP Stream.
    - Changed the stream view to raw data.
    - Saved the reconstructed file as employee_details.txt.
- Hash values (SHA-256 and MD5):
    - SHA-256: 77566e42d8a2d511a660a8b7d0641c3aeb62806fd7f881864fd0f0642aa3fa57
    - MD5: 33fe112801d0d5963ef014a25a8f731a

Kali Linux hash verification for recovered employee_details.txt

Recovered file contents included employee details and FTP credentials, including johnsmith/pass1234 and janedoe/1234abcd.

The recovered file contained employee personal information and plaintext FTP credentials. These credentials directly correspond to the later observed logins for the users johnsmith and janedoe in the packet capture.

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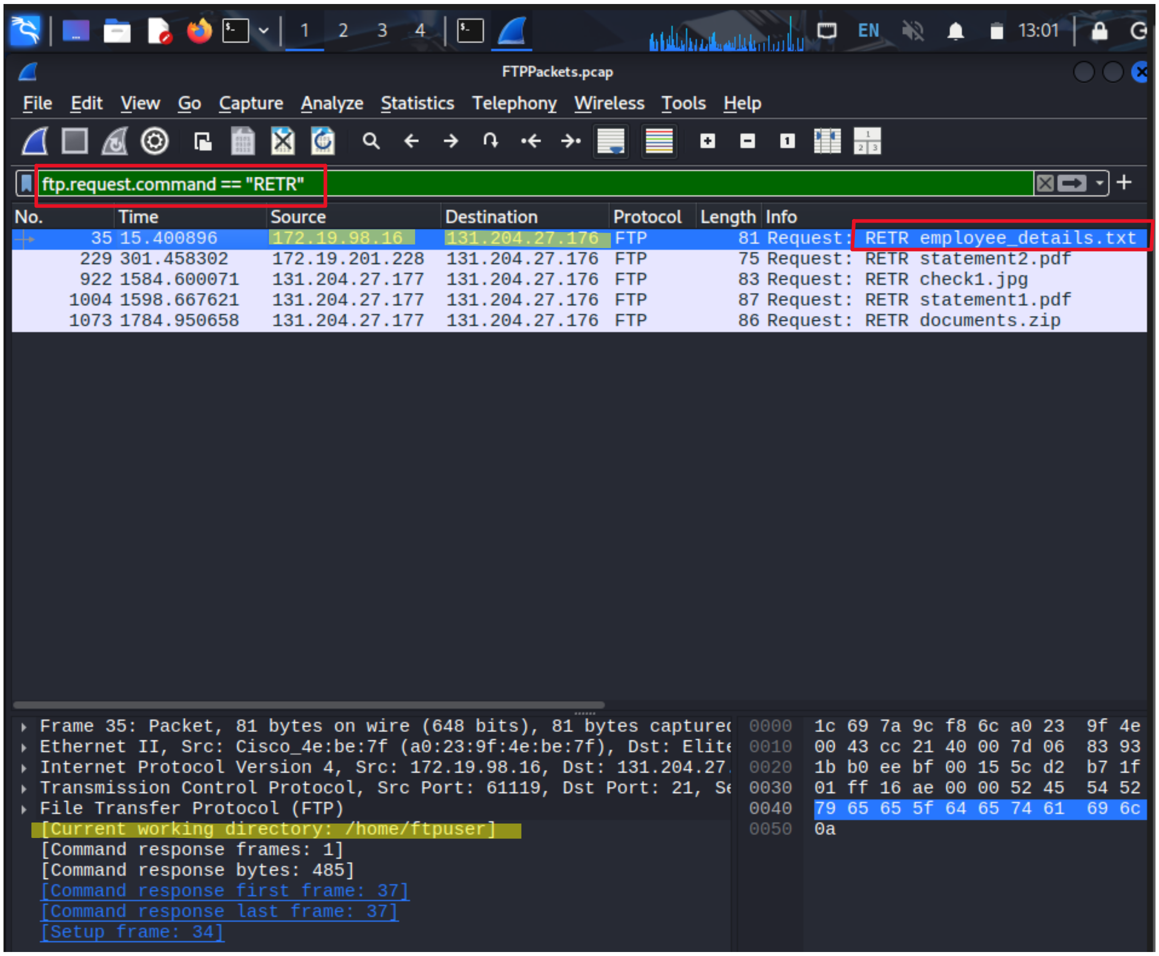

FTP RETR command showing the authorized user ftpuser requesting the file employee_details.txt from the FTP server 131.204.27.176

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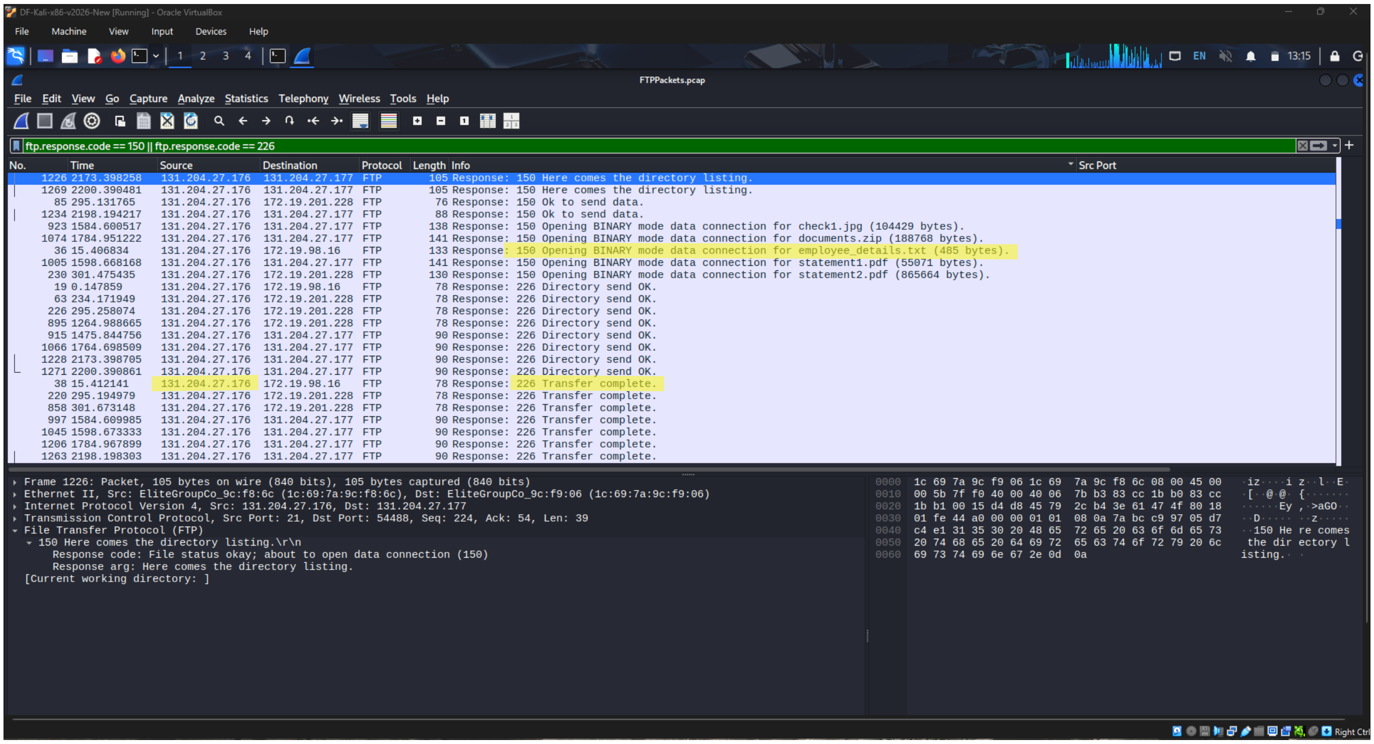

FTP response codes 150 and 226 confirming the successful transfer of employee_details.txt from the FTP server

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5. Unauthorized Access Enumeration

- Username: johnsmith
- IP address: 172.19.201.228
- MAC address (if observable): a0:23:9f:4e:be:7f
- Timestamp (DD-MM-YYYY HH:MM:SS): 15-11-2024 13:52:36
- FTP response codes: 331 (Username is okay, but needs password) and 230 (login successful)
- Login Attempt: Successful

- Username: janedoe
- IP address: 131.204.27.177
- MAC address (if observable): a0:23:9f:4e:be:7f
- Timestamp (DD-MM-YYYY HH:MM:SS): 15-11-2024 13:54:16
- FTP response codes: 331 (Username is okay, but needs password) and 230 (Login successful)
- Login Attempt: Successful

Username	Source IP	MAC Address	Login Timestamp	Assessment
ftpuser	172.19.98.16	a0:23:9f:4e:be:7f	15-11-2024 13:48:51	Authorized initial login
ftpuser	172.19.98.16	a0:23:9f:4e:be:7f	15-11-2024 14:02:15	Later reuse of authorized account
johnsmith	172.19.201.228	a0:23:9f:4e:be:7f	15-11-2024 13:52:55	Unauthorized – credentials exposed in employee_details.txt
johnsmith	172.19.201.228	a0:23:9f:4e:be:7f	15-11-2024 14:04:34	Unauthorized repeat login
janedoe	131.204.27.177	1c:69:7a:9c:f9:06	15-11-2024 13:56:40	Unauthorized – credentials exposed in employee_details.txt
janedoe	131.204.27.177	1c:69:7a:9c:f9:06	15-11-2024 14:18:06	Unauthorized repeat login

Both unauthorized accounts correspond to credentials contained within employee_details.txt, indicating that the credential file downloaded by ftpuser was likely used to authenticate subsequent unauthorized sessions.

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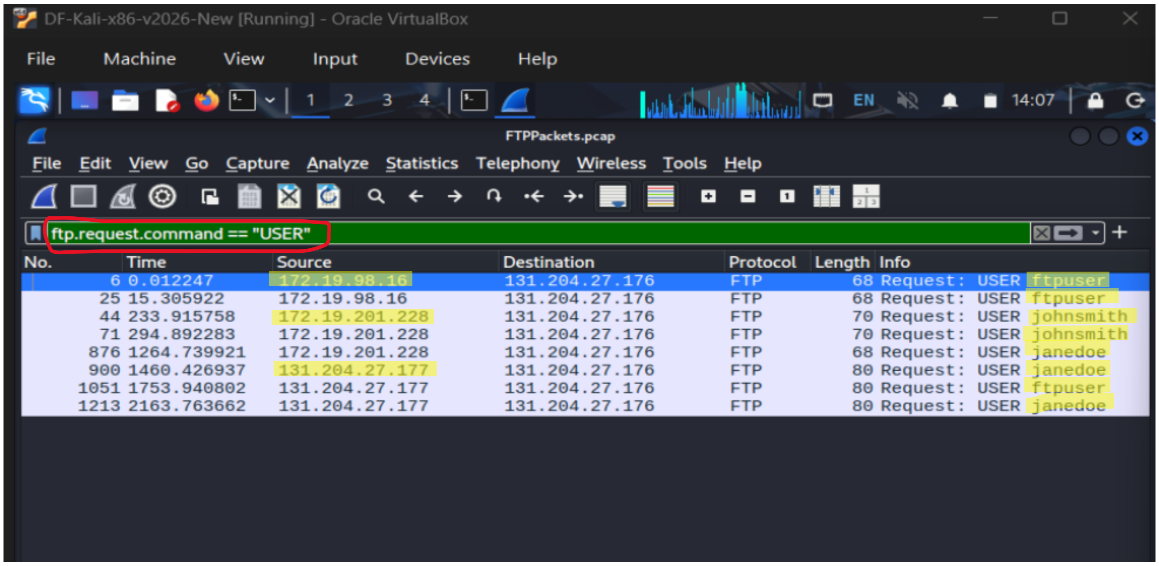

USER command filter showing all usernames observed in the FTP control channel above

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6. File Activity Per Login

Username	IP Address	Action	File Name	Size (bytes)	Integrity Notes
ftpuser	172.19.98.16	Download	employee_details.txt	485	Credential leak file containing usernames, passwords, and employee personal data
johnsmith	172.19.201.228	Upload	Check1.jpg	187392	Completed check image; differs from check1.jpg template
johnsmith	172.19.201.228	Download	Statement2.pdf	47104	Sample bank statement template
janedoe	131.204.27.177	Download	check1.jpg	104448	Sample/annotated check template
janedoe	131.204.27.177	Download	Statement1.pdf	55296	Realistic bank statement document
janedoe	131.204.27.177	Download	Documents.zip	189440	Archive containing identity-related images (SSN card and birth record)
janedoe	131.204.27.177	Upload	Mortgage.pdf	39936	Blank mortgage approval template

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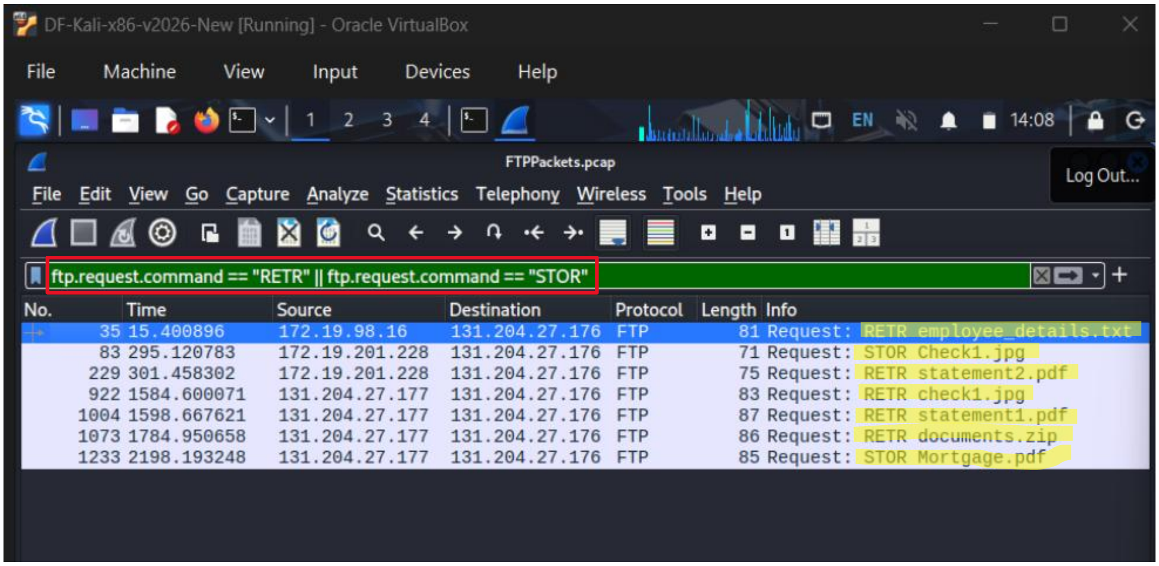

RETR/STOR filter showing all FTP file-transfer requests in the capture above

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7. Recovery of All Transferred Files

File Name	File Size (bytes)	SHA-256	MD5	Export Method
Employee_details .txt	485	77566e42d8a2d511a660a8b7d0641c3aeb62806fd7f881864fd0f0642aa3fa57	33fe112801d0d5963ef014a25a8f731a	Wireshark → Follow TCP Stream → Save Raw Data
Check1.jpg	187392	f87999f63685b4ddd0dc5193b2ce199c8eb7f0cf57b0bdb4c1924b53df7eed74	828a6aaf7d03425b709d1c934956e215	Wireshark → Export Objects → FTP
check1.jpg	104448	5031d7d419f8dc734ef0755238903b46bbcbf00148f20ac28f40ca2626032fc3	7127fbc910bba91881e4f92d9a60a3a4	Wireshark → Export Objects → FTP
Statement1.pdf	55296	6196dd97ba8071953c61e667dbdfaa1d6007025d85e150e75d29e82af9f241d9	5d73628bbc55be7a97b165793ff89a29	Wireshark → Export Objects → FTP
Statement2.pdf	47104	53cb0d6fcea0691c567f874309df843df7d7386842a54b3097825fc3ddeb235c	a347008364d3a9c640ee20fb098113c3	Wireshark → Export Objects → FTP
Documents.zip	189440	5cedf6a3089171e677adc7470f97b396d0f42168e95aa0925f0de5a0bf0ac722	7e256228ed332013dc8857bc4b735b88	Wireshark → Export Objects → FTP
Mortgage.pdf	39936	944393f110e37278dfeb31bb9bd250252e681593bfbd773a45bab48faa023fab7	1f4027a6add0e65c7ccd054a4fc066b	Wireshark → Export Objects → FTP

All reconstructed files listed above are in the submission zip file ‘recovered_files.zip’.

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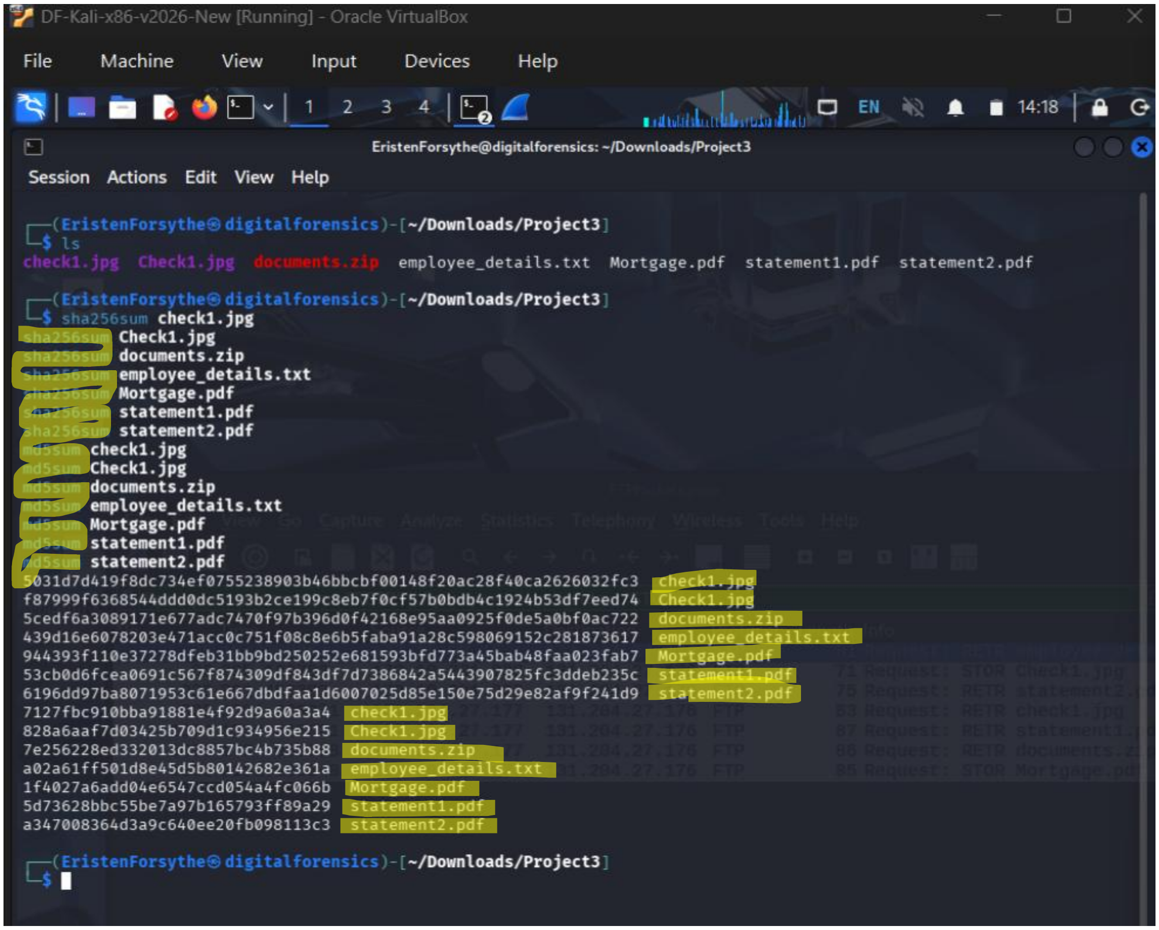
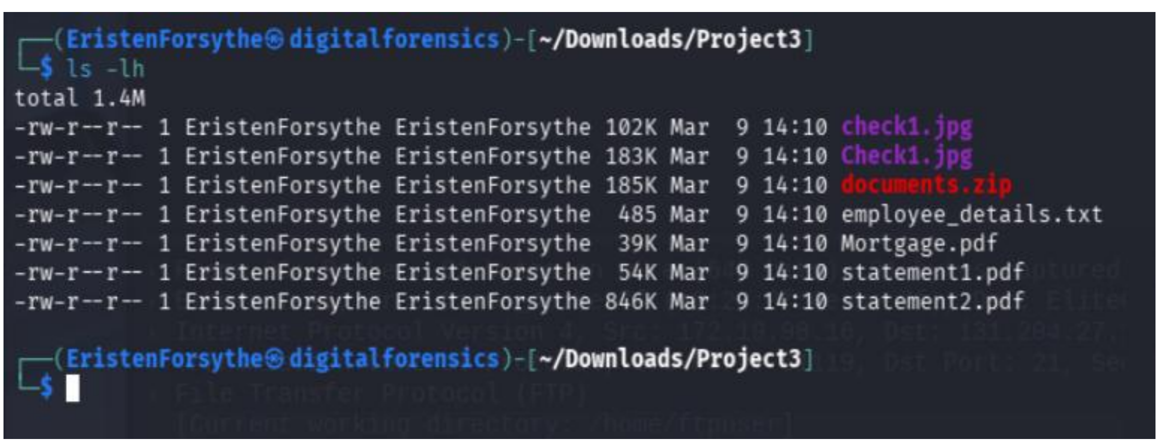


SHA-256 and MD5 hashes for all recovered files above



ls -lh output showing the recovered files in the Project3 directory

Files were reconstructed from the packet capture using two methods in Wireshark.

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The file employee_details.txt was recovered by locating the RETR command and using

Follow → TCP Stream to save the raw file data.

The remaining transferred files were recovered using

File → Export Objects → FTP in Wireshark, which reconstructs files transferred through the FTP data channel.

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8. Consolidated Findings Table (Required)

Username	IP Address	MAC Address	Login Timestamp (DD-MM-YYYY HH:MM:SS)	Action	File Name + Ext	File Size (Bytes)	File Hash
ftpuser	172.19.98.16	a0:23:9f:4e:be:7f	15-11-2024 13:48:51	Download	Employee_details.txt	485	SHA256: 77566e42d8a2d511a660a8b7d0641c3aeb62806fd7f881864fd0f0642aa3fa57
johnsmith	172.19.201.228	a0:23:9f:4e:be:7f	15-11-2024 13:52:36	Upload	Check1.jpg	187392	SHA256: f87999f63685b4ddd0dc5193b2ce199c8eb7f0cf57b0bdb4c1924b53df7eed74
johnsmith	172.19.201.228	a0:23:9f:4e:be:7f	15-11-2024 13:52:36	Download	Statement2.pdf	47104	SHA256: 53cb0d6fcea0691c567f874309df843df7d7386842a54b3097825fc3ddeb235c
janedoe	131.204.27.177	a0:23:9f:4e:be:7f	15-11-2024 13:54:16	Download	check1.jpg	104448	SHA256: 5031d7d419f8dc734ef0755238903b46bbcbf00148f20ac28f40ca2626032fc3
janedoe	131.204.27.177	a0:23:9f:4e:be:7f	15-11-2024 13:54:16	Download	Statement1.pdf	55296	SHA256: 6196dd97ba8071953c61e667dbdfaa1d6007025d85e150e75

							d29e82af9f241d9
janedoe	131.204.27.177	a0:23:9f:4e:be:7f	15-11-2024 13:54:16	Download	Documents.zip	189440	SHA256: 5cedf6a3089171e677adc7470f97b396d0f42168e95aa0925f0de5a0bf0ac722
janedoe	131.204.27.177	a0:23:9f:4e:be:7f	15-11-2024 13:54:16	Upload	Mortgage.pdf	39936	SHA256: 944393f110e37278dfeb31bb9bd250252e681593bfbd773a45bab48faa023fab7

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9. Conclusion

The analysis of the FTPPackets.pcap file showed that the authorized user ftpuser initially logged into the FTP server and downloaded the file employee_details.txt. This file had employee usernames, passwords, and personal information. Shortly after the download the credentials contained in the file were used to authenticate additional FTP sessions under the accounts johnsmith and janedoe.

Once authenticated, these accounts performed multiple file transfers involving images, PDFs, and a ZIP archive. The transferred files included financial documents such as bank statements, check images, and a mortgage template, as well as an archive containing identity related images like a social security card and birth certificate. All transferred files were successfully reconstructed from the packet capture and verified using SHA-256 and MD5 hashes.

The activity suggests that the exposed credentials enabled unauthorized access to the FTP server and allowed the attackers to exchange files containing personal identity information and financial documents. The materials could potentially be used for identity theft or financial fraud.

Overall the packet capture provided clear forensic evidence of credential exposure, unauthorized authentication, and subsequent file transfers. The recovered files and associated hash values allow the actions of each user session to be verified and reconstructed.